

LONDON SOUTHEND, United Kingdom

WMO: 036913

Lat: 51.571N

Lon: 0.696E

Elev: 49

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.9	29.8	21.3	16.0	32.4	23.3	17.6	32.5	29.9	49.8	27.2	49.8	4.5	280	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	80.4	66.3	76.9	64.9	73.8	62.8	67.7	77.4	65.9	74.3	64.4	72.1	9.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.5	72.0	62.6	85.1	70.5	61.0	80.2	68.7	32.1	77.1	30.7	74.3	29.6	72.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.7	19.6	22.3	86.9	3.2	2.8	20.0	88.9	18.1	90.5	16.2	92.1	13.9	94.2		
			21.3	72.0	2.9	2.6	19.2	73.9	17.4	75.4	15.8	76.8	13.6	78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	41.3	41.7	44.8	48.9	54.6	60.3	64.8	64.6	59.5	53.8	46.9	42.5
	DBStd	9.96	5.99	5.91	5.57	5.44	5.19	4.80	4.39	4.18	4.37	4.98	5.71	6.44
	HDD50	1176	272	236	177	83	18	0	0	0	0	21	125	244
	HDD65	4853	734	651	627	482	323	155	59	58	173	349	544	698
	CDD50	1919	3	5	15	51	161	309	458	452	285	138	31	11
	CDD65	119	0	0	0	0	1	13	52	45	7	1	0	0
	CDH74	669	0	0	0	1	9	91	287	248	32	1	0	0
	CDH80	113	0	0	0	0	1	13	46	50	3	0	0	0
Wind	WSAvg	9.1	10.3	10.5	9.7	8.6	8.7	8.6	8.6	8.5	8.4	9.0	9.1	9.7
Precipitation	PrecAvg	19.8	1.5	1.1	1.4	1.4	1.4	1.9	1.6	1.9	1.9	2.0	2.0	1.8
	PrecMax	25.9	4.8	2.4	3.3	2.6	2.6	4.3	4.1	4.3	6.5	5.1	4.0	3.7
	PrecMin	13.0	0.4	0.2	0.1	0.2	0.0	0.2	0.3	0.4	0.1	0.1	0.4	0.5
	PrecStd	2.9	0.8	0.6	0.9	0.7	0.7	1.1	0.9	0.9	1.5	1.3	0.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	57.5	61.2	69.9	75.0	81.0	84.5	85.8	77.4	69.5	61.2	57.5
		MCWB	51.7	51.7	52.2	57.8	62.8	67.2	67.2	68.2	64.7	61.6	57.1	53.4
	2%	DB	53.9	54.0	58.6	64.4	70.2	76.6	80.4	80.3	73.4	65.9	59.1	55.6
		MCWB	50.2	49.3	51.1	54.5	60.0	64.5	66.0	66.6	62.6	59.6	55.3	52.0
	5%	DB	52.0	52.1	55.6	60.9	67.6	73.1	77.0	76.6	69.9	62.9	57.3	53.9
		MCWB	48.5	47.7	49.4	52.6	58.3	62.2	64.7	65.1	61.2	58.0	53.3	50.4
	10%	DB	50.2	50.2	53.7	58.7	64.4	69.8	73.8	73.3	67.8	61.2	55.4	52.1
		MCWB	47.0	46.4	48.3	50.9	55.9	60.5	62.8	62.7	59.9	56.7	51.5	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.6	53.8	59.1	64.2	69.1	69.8	70.2	67.0	62.4	58.2	54.5
		MCDB	55.6	56.7	59.8	67.8	72.0	79.8	80.6	80.4	75.3	66.8	60.9	57.5
	2%	WB	51.3	50.6	52.1	55.6	61.0	65.4	67.5	67.9	64.3	60.3	55.6	52.8
		MCDB	54.0	54.1	56.7	62.6	68.7	74.2	77.4	77.5	70.7	64.0	58.6	55.7
	5%	WB	49.4	48.7	50.7	53.4	59.0	63.3	65.6	65.9	62.5	59.2	54.1	51.4
		MCDB	52.0	51.8	55.0	59.5	65.8	70.8	74.5	73.7	67.9	62.6	57.1	54.2
	10%	WB	47.5	47.0	49.0	51.5	56.9	61.1	64.1	64.3	60.9	57.5	52.1	49.4
		MCDB	50.2	50.1	53.3	57.1	62.9	68.0	71.9	71.5	65.9	60.9	55.2	52.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	11.7	14.3	14.7	14.9	15.4	15.1	13.9	11.8	9.7	8.6
		MCDBR	9.8	10.0	14.5	18.2	19.7	20.0	20.3	20.1	16.7	12.9	9.7	8.7
		MCWBR	9.2	8.0	9.8	10.9	11.7	11.1	9.9	9.8	9.3	9.3	8.3	8.0
	5% WB	MCDBR	9.9	9.6	13.3	17.2	18.1	18.5	18.4	18.1	14.9	12.0	9.6	8.6
		MCWBR	9.7	8.5	9.7	11.0	11.8	11.2	10.0	9.7	9.5	9.5	8.7	8.3
Clear-Sky Solar Irradiance	taub	0.332	0.349	0.386	0.407	0.413	0.414	0.411	0.402	0.381	0.365	0.337	0.323	
	taud	2.374	2.377	2.266	2.236	2.263	2.284	2.302	2.349	2.398	2.424	2.416	2.364	
	Ebn at Noon	213	244	254	263	267	267	266	262	254	234	209	195	
	Edh at Noon	20	26	34	40	41	40	39	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	246	459	814	1307	1555	1723	1636	1363	1023	586	298	191	
	RadStd	23	69	101	124	112	134	146	104	82	56	35	25	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page